Reinforcement Learning Project Proposal

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Abstract:

This project studies contextual bandit algorithms in an e-commerce recommendation setting using an open-source dataset collected from the fashion e-commerce platform ZOZOTOWN. The dataset contains large-scale logged bandit feedback, including contextual information, recommended items, observed rewards, and propensity scores generated by the logging policy.

Using this dataset, we investigate how different contextual bandit policies perform when making recommendation decisions based on user context. We also study off-policy evaluation (OPE) techniques that estimate the expected performance of new recommendation policies using historical interaction data without deploying them online. We will implement several bandit policies and compare multiple OPE estimators, including Inverse Propensity Weighting (IPW) and Doubly Robust methods.

Reference:

1. Taufiq, M. F., Doucet, A., Cornish, R., & Ton, J. F. (2023). Marginal density ratio for off-policy evaluation in contextual bandits. *Advances in Neural Information Processing Systems*, *36*, 52648-52691.
2. Saito, Y., Aihara, S., Matsutani, M., & Narita, Y. (2020). Open bandit dataset and pipeline: Towards realistic and reproducible off-policy evaluation. *arXiv preprint arXiv:2008.07146*.
3. Saito, Y., Ren, Q., & Joachims, T. (2023, July). Off-policy evaluation for large action spaces via conjunct effect modeling. In *international conference on Machine learning* (pp. 29734-29759). PMLR.
4. De Kerpel, L., & Benoit, D. (2022). The off-policy evaluation of Bayesian contextual bandit strategies in an e-commerce recommender system. In *32nd European Conference on Operational Research (EURO2022)*.

Direction & Outcomes:

In this project, we will investigate how contextual bandit algorithms perform in an e-commerce recommendation setting using offline data. We will implement several bandit policies, including ε-greedy, LinUCB, and Thompson Sampling, and apply off-policy evaluation methods such as IPW and Doubly Robust estimators to estimate policy performance from logged interaction data.

Through experiments on the dataset, we aim to compare the stability and accuracy of different OPE methods. The expected outcome is an empirical understanding of how contextual bandit policies and OPE estimators interact and which evaluation methods are more reliable for assessing recommendation policies in real-world scenarios.